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## Rotating device and electronic equipment accessory


### Abstract

A rotating device and an electronic equipment accessory are provided. The rotating device includes a first connecting member and a second connecting member rotatable about a rotation axis relative to the first connecting member. An angle formed between the rotation axis and a length direction of the rotating device is acute such that the second connecting member can move upwards when rotating with respect to the first connecting member. Therefore, when the rotating device is applied to an electronic equipment accessory which is mounted on electronic equipment, the rotating device acting as a stand can be arranged between a magnetic charging structure of electronic equipment and a bottom edge of the electronic equipment, such that the rotating device will not shield the magnetic charging structure. The rotating device can act as a stand with a good support effect due to existing of the acute angle.



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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present disclosure is a continuation of International Patent Application No. PCT/CN2023/132148 filed on Nov. 16, 2023, which claims priorities of China patent applications No. 202223095893.1, filed on Nov. 16, 2022, No. 202322914218.5 filed on Oct. 26, 2023 and No. 202322414061.X filed on Sep. 5, 2023. The entire contents of which are incorporated herein by reference.

FIELD
(1) The application relates to the technical field of portable electronic equipment accessories, in particular to a rotating device and an electronic equipment accessory.
BACKGROUND
(2) Existing electronic equipment accessories with support functionality, such as mobile phone cases, usually have a stand which is connected to a phone case via a rotating device. In order to enable both horizontally and vertically support for the electronic equipment such as a mobile phone, the stand is usually positioned closer to the middle area of the electronic equipment accessory when in use.
(3) However, the middle areas of current mobile phones are usually provided with functional components like magnetic attraction structure and wireless charging coils. To avoid interference with the use of the magnetic attraction structure and wireless charging coils, the stands of electronic device accessories generally need to be positioned away from the middle areas of the mobile phones. This means that the stands of the electronic device accessories usually need to be set near the edges of the mobile phones. But when the stands are set near the edges of the mobile phones, the stands cannot support the mobile phones stably.
SUMMARY
(4) To ensure that an electronic equipment stand can realize a support function and keep away from the magnetic charging structure of electronic equipment, the embodiments of the application provide a rotating device and an electronic equipment accessory.
(5) In one aspect, the present application provides a rotating device comprising a first connecting member and a second connecting member. The second connecting member is rotatable about a rotation axis relative to the first connecting member, and an angle formed between the rotation axis and a length direction of the rotating device is acute.
(6) In some embodiments, the angle is greater than or equal to 40° and less than or equal to 65°.
(7) In some embodiments, the rotating device further comprises a shaft, wherein the second connecting member and the first connecting member are rotatably connected to each other via the shaft; the first connecting member is configured to connect with a mobile phone protective case; and the second connecting member is configured to support the mobile phone protective case directly or support the mobile phone protective case via a support member.
(8) In some embodiments, a maximum thickness of the rotating device in a thickness direction is greater than or equal to 2 mm and less than or equal to 4 mm, the thickness direction being

perpendicular to the length direction and the rotation axis.
(9) In another aspect, the present invention provides an electronic equipment accessory which comprises an accessory body comprising a back panel, the back panel having a top edge and a bottom edge opposite to the top edge, the back panel comprising a first area adjacent to the bottom edge; and a rotating device mounted to the back panel and located at the first area, the rotating device comprising a shaft which inclines with respect to the bottom edge.
(10) In some embodiments, an angle formed between an axial direction of the shaft and a length direction of the bottom edge is greater than or equal to 45° and less than or equal to 75°.
(11) In some embodiments, the rotating device comprises a first connecting member and a second connecting member which are pivotably connected to each other via the shaft; and the back panel defines a through opening in which the shaft is located.
(12) In some embodiments, the back panel comprises an inner surface and an outer surface opposite to the inner surface in a thickness direction of the back panel; a receiving groove is defined in the outer surface for receiving the second connecting member; and the first connecting member is at least partly fixed to the inner surface of the back panel.
(13) In some embodiments, the receiving groove comprises a first bottom wall and a second bottom wall offset from the first bottom wall in the thickness direction of the back panel, a through hole is defined between the first bottom wall and the second bottom wall, and the first connecting member passes through the through hole.
(14) In some embodiments, the first connecting member comprises a first connection portion, a bent portion and an extension portion arranged in sequence; the extension portion is located at the outer surface of the back panel; the first connection portion is located at the inner surface of the back panel; and the bent portion is located at the through hole.
(15) In some embodiments, the electronic equipment accessory further comprises a first attachment part which is located at the receiving groove and covers the first connecting member.
(16) In some embodiments, the electronic equipment accessory further comprises an attachment part which is located at the inner surface of the back panel opposite to the receiving groove; and the attachment part covers the first connecting member and the through opening.
(17) In some embodiments, the receiving groove comprises a first bottom wall and a second bottom wall offset from the first bottom wall in the thickness direction of the back panel; a through hole is defined between the first bottom wall and the second bottom wall; and the first bottom wall is sandwiched between the first connecting member and a mounting member in the thickness direction of the back panel.
(18) In some embodiments, a support member is connected to the second connecting member and rotatable together with the second connecting member relative to the first connecting member, and the first connecting member is fixedly connected to the back panel; the support member is connected to a side of the second connecting member opposite to the back panel, the second connecting member is connected to the shaft.
(19) In some embodiments, the second connecting member acts as or is connected to a support member; and the support member comprises an arcuate outer end opposite to the shaft; the receiving groove comprises an arcuate inner end wall opposite to the shaft; the receiving groove defines a length direction and a width direction, and the arcuate inner end wall of the receiving groove faces the arcuate outer end of the support member with a gap formed therebetween; and a size of the gap in the length direction gradually decreases from a middle thereof to opposite sides thereof in the width direction.
(20) In some embodiments, the back panel comprises a second area which is located between the first area and the top edge, and a magnetic attraction component is disposed at the second area.
(21) In some embodiments, the magnetic attraction component comprises a first magnetic attraction member which comprises a plurality of segmented magnets arranged in a ring shape; and the back panel is provided with a ring-shaped cover stacked on the first magnetic attraction member in a

thickness direction of the back panel.
(22) In some embodiments, the magnetic attraction component comprises a second magnetic attraction member which has a strip shape and is located between the first magnetic attraction member and the rotating device; and the back panel is provided with a strip-shaped cover stacked on the second magnetic attraction member in the thickness direction of the back panel.
(23) In some embodiments, the second connecting member acts as or is connected to a support member; the support member comprises a support point located at a free end thereof. When the free end of the support member is moved away from the accessory body to be located at a support state, the support point and the bottom edge cooperatively define a support surface and an included angle formed between the support surface and the back panel is greater than or equal to 60° and less than or equal to 75°.
(24) In some embodiments, one side of the mounting member facing the second connecting member defines a notch, and the shaft is received in the notch.
(25) The above technical solution of the application has at least the following beneficial effects: when the rotating device is applied to an electronic equipment accessory which is mounted on electronic equipment and used as a stand of the electronic equipment, the rotating device can be arranged between a magnetic charging structure of the electronic equipment and a bottom edge of the electronic equipment, such that the rotating device will not shield the magnetic charging structure. The rotating device can act as a stand with a good support effect due to existing of the acute angle.

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## Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) To more clearly explain the technical solutions of the embodiments of the application, drawings used for describing the embodiments will be briefly introduced below. Obviously, the drawings in the following description merely illustrate some embodiments of the application, and those ordinarily skilled in the art can obtain other drawings according to the following ones without creative labor.
(2) FIG. **1** is a perspective structural view of a rotating device according to Embodiment 1 of the application.
(3) FIG. **2** is a top plan view of the rotating device in FIG. **1**.
(4) FIG. **3** is a side view of the rotating device in FIG. **1**.
(5) FIG. **4** illustrates dimensions of the rotating device in FIG. **3**.
(6) FIG. **5** is an exploded view of the rotating device in FIG. **1**.
(7) FIG. **6** is a perspective view of an electronic equipment accessory according to Embodiment 2 of the application.
(8) FIG. **7** is a front view of the electronic equipment accessory in FIG. **6**.
(9) FIG. **8** is an exploded view of the electronic equipment accessory in FIG. **6**.
(10) FIG. **9** is a perspective view of an electronic equipment accessory according to Embodiment 3 of the application.
(11) FIG. **10** is a partial sectional view of a spindle structure of the electronic equipment accessory in FIG. **9**.
(12) FIG. **11** is an exploded view of the electronic equipment accessory in FIG. **9**.
(13) FIG. **12** is a perspective view of another electronic equipment accessory according to Embodiment 3 of the application.
(14) FIG. **13** is a perspective view of the electronic equipment accessory in FIG. **12**, in another state.
(15) FIG. **14** is a sectional view of a spindle structure of the electronic equipment accessory in FIG.

**12**.
(16) FIG. **15** is an exploded view of the electronic equipment accessory in FIG. **12**.
(17) FIG. **16** is a back view of an electronic equipment accessory according to Embodiment 4 of the application.
(18) FIG. **17** is a perspective view of the electronic equipment accessory in FIG. **16**, from another perspective.
(19) FIG. **18** is a front view of the electronic equipment accessory in FIG. **16**.
(20) FIG. **19** is an exploded view of the electronic equipment accessory in FIG. **16**, from another perspective.
(21) FIG. **20** is a perspective view of the electronic equipment accessory in FIG. **16** in a case where a stand is in a folded state.
(22) FIG. **21** is a perspective view of the electronic equipment accessory in FIG. **20** in a case where the stand is in a support state.
(23) FIG. **22** is a side view of the electronic equipment accessory in FIG. **21** in a case where the electronic equipment accessory is supported on a support surface by means of the stand.
(24) FIG. **23** is an enlarged and exploded perspective structural view of the stand in the electronic equipment accessory in FIG. **20**.
(25) FIG. **24** is an enlarged view in a case where the stand in FIG. **23** is in a folded state.
(26) FIG. **25** is an enlarged perspective structural view in a case where the stand in FIG. **24** is in a support state.
(27) FIG. **26** is an exploded perspective structural view of the electronic equipment accessory in FIG. **17**.
(28) FIG. **27** illustrates an electronic equipment accessory according to Embodiment 5 of the application.
(29) FIG. **28** is an exploded view of the electronic equipment accessory in FIG. **27**.
(30) FIG. **29** is a sectional view of the electronic equipment accessory in FIG. **27**, wherein a rotating device of the electronic equipment accessory is in an unfolded state.
(31) FIG. **30** is a schematic view in case where a rotating device of the electronic equipment accessory in FIG. **27** is in an unfolded state.
(32) FIG. **31** is a schematic view in a case where the rotating device in FIG. **30** is in a folded state.
(33) FIG. **32** is a schematic view of the electronic equipment accessory in FIG. **27** in a vertically supported state.
(34) FIG. **33** is a schematic view of the electronic equipment accessory in FIG. **27** wherein the rotating device of the electronic equipment accessory is in an unfolded state.
(35) FIG. **34** is a schematic view of the electronic equipment accessory in FIG. **27** in a horizontally supported state.

DESCRIPTION OF THE EMBODIMENTS

(36) The technical solutions in the embodiments of the application will be clearly and completely described below in conjunction with the drawings of the embodiments of the application. Obviously, the embodiments in the following description are merely illustrative ones and are not all possible ones of the application. All other embodiments obtained by those ordinarily skilled in the art based on the following ones without creative labor should also fall within the protection scope of the application.
(37) It should be noted that terms such as “first”, “second”, “one end” in the description and claims of the application are used for distinguishing similar objects rather than describing a specific sequence or precedence order. It should be understood that these terms can be exchanged in an appropriate case to allow the embodiments of the application described here to be implemented in other sequences different from those illustrated or described here. In addition, terms such as “comprise” and “have” and any variants thereof are intended to indicate non-exclusive inclusion. For example, a process, system, product or device comprising a series of steps or units should not

be limited to the steps or units that are clearly listed, and may also comprise other steps or units that are not clearly listed or other inherent steps or units of the process, product or device.

Embodiment 1

(38) Referring to FIGS. **1** and **2**, Embodiment 1 of the application provides a rotating device **1**. In this embodiment, the rotating device **1** is described with a hinge device as an example. The rotating device **1** comprises a first connecting member **11**, a second connecting member **12** and a shaft **13**. The first connecting member **11** and the second connecting member **12** are rotatably connected by means of the shaft **13**. The first connecting member **11** and the second connecting member **12** extend respectively in a length direction x of the rotating device **1**. The second connecting member **12** is rotatable relative to the first connecting member **11** about a rotation axis z. A preset acute angle A is formed between the rotation axis z and the length direction x. In this embodiment, the rotation axis z coincides with the axis of the shaft **13**.

(39) According to the rotating device **1** provided by Embodiment 1 of the application, by means of the preset acute angle A formed between the rotation axis z and the length direction x, the second connecting member **12** can move upwards as shown in FIG. **2** when rotating with respect to the first connecting member **11** about the rotation axis z shown in FIG. **2**. In this way, when the rotating device **1** is applied to an electronic equipment stand, the first connecting member **11** and the second connecting member **12** can be respectively connected to support members extending in the length direction x, and the second connecting member **12** can drive the corresponding support member connected thereto to turn upwards to allow the electronic equipment stand to be disposed at a position between a magnetic charging structure of electronic equipment and a bottom edge of the electronic equipment, such that the electronic equipment stand will not shield/interfere the magnetic charging structure. In addition, because of existing of the preset acute angle A of the rotating device **1**, the electronic equipment can be well supported by the electronic equipment stand. Of course, the rotating device **1** may also be mounted on an electronic equipment accessory such as an electronic equipment protective case and be turned away from an area corresponding to the magnetic charging structure on the electronic equipment protective case. In this embodiment, the rotation axis z is defined by the axis of the shaft **13**.

(40) Specifically, the first connecting member **11** and the second connecting member **12** may be rotatably connected to each other by means of a rod structure and a hole structure respectively formed on the first connecting member **11** and the second connecting member **12**. For example, a connecting rod acting as a shaft may be arranged on one side of the second connecting member **12**, a connecting hole is formed in one side of the first connecting member **11**, and the connecting rod of the second connecting member **12** is pivotably received in the connecting hole of the first connecting member **11**, such that the first connecting member **11** and the second connecting member **12** are rotatably connected to each other. Alternatively, the first connecting member **11** and the second connecting member **12** are respectively sleeved around the shaft **13** to be rotatably connected to each other. Or, one of the first connecting member **11** and the second connecting member **12** is fixedly connected to the shaft **13**, and the other one of the first connecting member **11** and the second connecting member **12** is rotatably connected to the shaft **13**.

(41) Specifically, the electronic equipment may be a smartphone, a tablet, a portable fan, or the like. In actual use, any one of or both the first connecting member **11** and the second connecting member **12** may be connected to a support member and may be turnable/foldable to fulfil a supporting function together with the support member. Any one of the first connecting member **11** and the second connecting member **12** may be only used for connection and fixation. For example, the first connecting member **11** is fixed to the electronic equipment or the electronic equipment accessory, wherein the electronic equipment accessory is, for example, an electronic equipment stand, a wireless charger, a portable power supply, a bag or card case, or an electronic equipment protective case. It should be noted that when one of the first connecting member **11** and the second connecting member **12** is fixedly connected to the electronic equipment or the electronic equipment

accessory, the other one of the first connecting member **11** and the second connecting member **12** can be used directly as a support component, that is, no extra support member is needed. For example, the second connecting member **12** acts as one support member and is connected to the electronic equipment protective case or portable power supply via the first connecting member **11** to thereby provide a support function for the electronic equipment protective case or portable power supply.

(42) Preferably, as shown in FIG. **2**, the preset acute angle A is greater than or equal to 40° and less than or equal to 65°. In this way, when the rotating device **1** is applied to an electronic equipment stand, the electronic equipment stand can fulfil a good support effect for the electronic equipment.

(43) Referring to FIGS. **3** and **4**, the first connecting member **11** comprises a first connection portion **111**, the second connecting member **12** comprises a second connection portion **121**, and the first connection portion **111** and the second connection portion **121** are arranged on opposite sides of the shaft **13** respectively. The first connection portion **111** and the second connection portion **121** are offset from each other in a thickness direction y of the rotating device **1**. The first connection portion **111** has a first connection surface **1111**, and the second connection portion **121** has a second connection surface **1211**.

(44) In a case where the rotating device **1** is applied to an electronic equipment stand, due to the first connection portion **111** and the second connection portion **121** are offset in the thickness direction y of the rotating device **1**, when the electronic equipment stand is mounted on, for example, an electronic equipment protective case, the first connecting member **11** and the second connecting member **12** may be respectively mounted on opposite sides of a sidewall of the electronic equipment protective case. For example, as shown in FIG. **3**, one portion of the sidewall is located above the first connection surface **1111**, and the other portion of the sidewall is located at the other side of the second connection portion **121** opposite to the second connection surface **1211**, such that the connection stability of the rotating device **1** and the electronic equipment protective case is improved. In addition, by means of the offset arrangement between the first and second connecting members **11**/**12**, a rotary joint between the first connecting member **11** and the second connecting member **12** is located at the same height as one portion of the sidewall in the thickness direction y, such that the rotary joint can be hidden in the electronic equipment protective case to some extent, and the thickness of the electronic equipment protective case does not need to be increased. Specifically, the misalignment of the first connection portion **111** and the second connection portion **121** may be defined by means of a height difference D**1**.

(45) Furthermore, referring to FIG. **3**, the first connection surface **1111** and the second connection surface **1211** are arranged on the same side of the rotating device **1** in the thickness direction, and the height difference D**1** is formed between the first connection surface **1111** and the second connection surface **1211**.

(46) The height difference D**1** is defined by the first connection surface **1111** and the second connection surface **1211**, so when the rotating device **1** is connected to the electronic equipment protective case, for example, to a back panel of the electronic equipment protective case, one portion of the back panel may be located above the upper side of the first connection surface **1111** and the other portion of the back panel may be located below the lower side of the second connection surface **1211**. In this way, one portion of the rotating device **1** may be hidden below the back panel and the second connecting member **12** may be located above the back panel, thus reducing the height of the rotating device **1** protruding out of the back panel. The first connecting member **11** is hidden below the back panel while the second connecting member **12** is located above the back panel, which makes the connection between the rotating device **1** and the back panel more stable.

(47) Specifically, the definition of the height difference D**1** by means of the first connection surface **1111** and the second connection surface **1211** is only one approach for defining the height difference D**1** provided by this embodiment. The height difference D**1** may also be defined by

center lines of the first connection portion **111** and the second connection portion **121** in the thickness direction y or by the distance between two adjacent surfaces of the first connection portion **111** and the second connection portion **121** in the thickness direction y. In addition, the thickness of the first connection portion **111** and the thickness of the second connection portion **121** in the thickness direction y may be equal or unequal, so the thickness of the first connection portion **111** may be different from the thickness of the second connection portion **121** in some cases. In a case where the thickness of the first connection portion **111** is different from the thickness of the second connection portion **121**, those skilled in the art need to set the position of the first connection portion **111** and the position of the second connection portion **121** in the thickness direction y according to the actual outlines (including the thicknesses) of the first connection portion **111** and the second connection portion **121** to realize the misaligned arrangement.
(48) Further, as shown in FIG. **4**, a maximum thickness D**2** of the rotating device **1** in the thickness direction y is greater than or equal to 2 mm and less than or equal to 4 mm. Due to the distance between a wireless charger using an electromagnetic structure for wireless charging and the coil of the electronic equipment should not be too long, the thickness of the back panel of most electronic equipment protective cases on the present market is 2-4 mm, so the maximum thickness D**2** of the rotating device **1** is set to 2-4 mm to ensure that the shaft will not protrude out of the back panel in the thickness direction y when the rotating device **1** is mounted on the back panel. In this way, the surface of the back panel of the electronic equipment protective case is kept smooth, thus improving the sense of touch and aesthetics of the electronic equipment protective case when the electronic equipment protective case is held.
(49) Specifically, the rotating device **1**, for example, is configured as a sheet structure on the whole and thus has the length direction x and the thickness direction y shown in FIGS. **2-4**. Referring to FIG. **4**, the thickness D**4** of the first connecting member **11** and the second connecting member **12** may be set to, for example, 0.5 mm. Correspondingly, the thickness of the shaft **13** in the thickness direction y is set to 1-3 mm.
(50) Furthermore, referring to FIGS. **2-5**, the shaft **13** has a shaft stop surface **131**. The first connecting member **11** has a first rotary connection portion **112**. The first rotary connection portion **112** is disposed around the shaft **13** and has a first stop surface **1121** corresponding to the shaft stop surface **131**. The shaft stop surface **131** works together with the first stop surface **1121** to limit a rotation position of the first connecting member **11** with respect to the shaft **13**, and the shaft stop surface **131** can also increase the rotational damping force of the first connecting member **11** with respect to the shaft **13**.
(51) Specifically, the shaft stop surface **131** and the first stop surface **1121** extend, for example, in the rotation axis direction z. The shaft stop surface **131** and the first stop surface **1121** may be flat surfaces, curved surfaces, or combinations of multiple flat surfaces and curved surfaces. Multiple shaft stop surfaces **131** may be arranged at intervals, and multiple first stop surfaces **1121** may be arranged corresponding to the multiple shaft stop surfaces **131**. The first rotary connection portion **112** is sleeved on the shaft **13**, thus being in contact with the circumferential surface of the shaft **13**. For example, in the state shown in FIGS. **3** and **4**, the first stop surface **1121** abuts against the shaft stop surface **131**, and in this state, the first stop surface **1121** and the shaft stop surface **131** work together to limit the rotation angle of the first connecting member **11** with respect to the shaft **13**. Furthermore, by means of the first stop surface **1121** and the shaft stop surface **131**, the second connecting member **12** can automatically restore when it is rotated to a certain angle relative to the first connecting member **11**. The cross-section, orthogonal to the rotation axis direction z, of the shaft **13** may be a circular cross-section, an oval cross-section, a section formed by a closed curve, a cross-section defined by a polygon, or a cross-section defined by a variable number of curved segments and straight segments. The shape of the cross-section of the shaft **13** depends on the arrangement of a circumferential surface of the shaft **13** in the rotation axis direction z. For example, the shaft stop surface **131** may be an uneven curved surface, so when the first connecting

member **11** rotates around the shaft **13**, the shaft stop surface **131** can increase the damping force on the first connecting member **11** during rotation. On this basis, the first stop surface **1121** corresponding to the shaft stop surface **131** and arranged on the first connecting member **11** can further increase the damping force and limit the amplitude of rotation of the first connecting member **11** with respect to the shaft **13**. Specifically, the first stop surface **1121** corresponds to the shaft stop surface **131**, which means that, for example, the shaft stop surface **131** and the first stop surface **1121** are in contact with each other, meshed with each other, or in other connection relations that can fulfil a damping and/or limiting effect.

(52) For example, the shaft stop surface **131** may be an uneven curved surface, and the projection of the curved surface in a plane perpendicular to the axis of the shaft **13** is approximately circular or oval. Correspondingly, the first stop surface **1121** may be in a corresponding shape. Or, the shaft stop surface **131** is formed by multiple flat surfaces arranged in the circumferential direction of the shaft **13**, such that the shaft **3** has a quadrangular prism shape, a pentagonal prism shape or other shapes of multi-sided prisms. The first rotary connection portion **112** is sleeved on the shaft **13**, so the inner surface of the connecting hole of the first rotary connection portion **112** acts as the first stop surface **1121**; and the connecting hole may be configured as a cavity matching the shaft with a prism shape, such that the first stop surface **1121** corresponds to and conforms with the shaft stop surface **131**.

(53) Furthermore, referring to FIGS. **3-5**, two shaft stop surfaces **131** are arranged and appositely located on opposite sides of the shaft **13**. Two first stop surfaces **1121** are arranged in one-to-one correspondence with the two shaft stop surfaces **131**.

(54) In some embodiments, multiple shaft stop surfaces **131** are arranged at intervals, such that the distance between the edge of the cross-section, perpendicular to the rotation axis direction z, of the shaft **13** and the rotation center of the shaft **13** is variable, as shown in FIGS. **3** and **4**. Specifically, the length of the shaft **13** in the length direction x is greater than the thickness of the shaft **13** in the thickness direction y. In this way, after rotating clockwise around the rotation axis direction z shown in FIG. **3** or **4**, the first rotary connection portion **112** will deform due to a dimension variation of the shaft **13** in different radial directions, thus increasing the rotational damping force of the first rotary connection portion **112** and being capable of maintaining the first rotary connection portion **112** at position after the first rotary connection portion **112** stops rotation. In addition, the shaft stop surface **131** and the first stop surface **1121** are in contact with each other in the state shown in FIGS. **3** and **4**, which facilitates to prevent the first connecting member **11** from rotating randomly due to loose connection between the first connecting member **11** and the shaft **13**. In this embodiment, only the second connecting member **12** is allowable to rotate relative to the shaft **13**.

(55) Further, referring to FIGS. **3-5**, the second connecting member **12** has a second rotary connection portion **122**. The second rotary connection portion **122** is disposed around the shaft **13** and has the second stop surface **1221** corresponding to the shaft stop surface **131**. The second stop surfaces **1221** may be in contact with the shaft stop surfaces **131**. The technical effect fulfilled by the second stop surface **1221** and the shaft stop surface **131** can be understood with reference to the technical effect fulfilled by the first stop surface **1121** and the shaft stop surface **131**. That is, the second stop surfaces **1221** is in contact with the shaft stop surfaces **131**, which can increase the rotational damping force of the first rotary connection portion **112** and maintain the first rotary connection portion **112** at position after the first rotary connection portion **112** is rotated to a desired angle with respect to the shaft **13**. Specifically, structure and shape of the second stop surface **1221** may be designed with reference to that of the first stop surface **1121**.

Embodiment 2

(56) Referring to FIGS. **6-8**, an electronic equipment accessory is provided according to Embodiment 2 of the application. In this embodiment, the electronic equipment accessory is described with an electronic equipment stand as an example. The electronic equipment accessory

comprises the rotating device **1** provided by Embodiment 1 of the application, a mounting member **14** and a support member **15**. The mounting member **14** is connected to the first connecting member **11** in order to connect the first connecting member **11** to an electronic equipment or an electronic equipment protective case. The support member **15** is connected to the second connecting member **12** to form a stand for supporting the electronic equipment. One side of the mounting member **14** facing the second connecting member **12** defines a notch **140**. The shaft **13** is received in the notch **140**. The use, implementation and beneficial effects of the electronic equipment accessory can be understood with reference to Embodiment 1 of the application.

(57) In some embodiments, the mounting member **14** and the first connecting member **11** may be formed integrally, and the support member **15** and the second connecting member **12** may be formed integrally. That is, the first connecting member **11** and the second connecting member **12** are designed according to the shape and size of an electronic equipment stand, thereby making the rotating device **1** directly form the electronic equipment stand.

Embodiment 3

(58) Referring to FIGS. **9-11**, Embodiment 3 of the application provides an electronic equipment accessory. The electronic equipment accessory comprises: a back panel **21**, the rotating device **1** provided by Embodiment 1 of the application, a mounting member **14** and a support member **15**. The back panel **21** defines a receiving groove **211**. The mounting member **14** is connected to the first connecting member **11**. The support member **15** is connected to the second connecting member **12**. The receiving groove **211** is configured for receiving the rotating device **1**, the mounting member **14** and the support member **15**.

(59) Specifically, a receiving cavity **212** is defined in one side of the electronic equipment accessory. The electronic equipment accessory, for example, further comprises a frame **22** which surrounds and is connected to the back panel **21** to cooperatively form the receiving cavity **212**. The receiving groove **211**, for example, is arranged on a side, away from the receiving cavity **212**, of the back panel **21**. The electronic equipment accessory, for example, may be an electronic equipment protective case, an electronic equipment stand, a portable power supply, a wireless charger, a bag or card holder, or the like.

(60) Further, referring to FIGS. **10** and **11**, the receiving groove **211** comprises a first bottom wall **213** and a second bottom wall **214** which are sunken inwardly relative to the outer surface of the back panel **21** in the thickness direction of the back panel **21**. A through-opening **215** is formed between the first bottom wall **213** and the second bottom wall **214**, and the rotating device **1** passes through the through-opening **215**. The first bottom wall **213** is sandwiched between the first connecting member **11** and the mounting member **14**, and the first connecting member **11** is arranged on a side, away from the receiving groove **211**, of the first bottom wall **213**, and the second connecting member **12** is arranged in the receiving groove **211**.

(61) Furthermore, referring to FIGS. **3** and **10**, the first connection surface **1111** is in contact with the side, away from the receiving groove **211**, of the first bottom wall **213**. The second connection portion **121** is arranged in the receiving groove **211**, and the second connection surface **1211** is located on a side, away from the second bottom wall **214**, of the second connection portion **121**.

(62) Further, referring to FIGS. **7-11**, the electronic equipment accessory further comprises a second attachment part **25**, which corresponds to the receiving groove **211** in the receiving cavity **212**. Specifically, the second attachment part **25** is, for example, an adhesive patch, which is adhered to a surface, opposite to the first connection surface **1111**, of the first connection portion **111**. The second attachment part **25** is configured for covering and protecting the rotating device **1**.

(63) Therefore, when the electronic equipment accessory is provided with the rotating device **1** shown in FIGS. **3** and **4**, a layered structure shown in FIG. **10** may be formed. Wherein, the first connecting member **11** is arranged on a lower side of the first bottom wall **213**, and the second connecting member **12** is arranged on an upper side of the second bottom wall **214**. In this way, the connection stability of the rotating device **1** and the electronic equipment accessory is improved, a

rotary joint between the mounting member **14** and the support member **15** can be hidden in the back panel **21**, and the thickness of the back panel **21** does not need to be increased. The support member **15** is rotatable with respect to the electronic equipment accessory between an extended position and a closed position. Preferably, at the closed position, the support member **15** is fully received in the receiving groove **211**, and the support member **15** and the back panel **21** are coplanar. At the extended position, the support member **15** is rotated out of the receiving groove **211** and can cooperate with the frame of the electronic equipment accessory to support an electronic equipment on a support surface.

(64) Referring to FIGS. **12-15**, Embodiment 3 of the application further provides an alternative electronic equipment accessory with a different appearance. The outline of the electronic equipment accessory shown in FIGS. **12-15** is different from the outline of the electronic equipment accessory shown in FIGS. **9-11**. The electronic equipment accessory shown in FIGS. **12-15** may be a portable power supply or a protective case configured for protecting a portable power supply such as a power bank, and the electronic equipment accessory shown in FIGS. **9-11** may be used for protecting electronic equipment such as a smartphone or a tablet.

(65) Further, referring to FIG. **14**, the mounting member **14** comprises an extension part **141** which extends towards the support member **15** and stretches across the shaft **13**. The second connecting member **12** has a bent portion **123** on a side close to the shaft **13**, and the bent portion **123** bends towards a side away from the extension part **141**. The extension part **141** may cover the shaft **13**, a connection structure between the first rotating member **11** and the shaft **13**, as well as a connection structure between the second connecting member **12** and the shaft **13**, such that the shaft **13** is prevented from being exposed to the outside of the mounting member **14** and the support member **15**. To ensure that the mounting member **14** and the support member **15** can rotate relative to each other smoothly, the bent portion **123** of the second rotating member **12** may keep away from the extension part **141**.

Embodiment 4

(66) Referring to FIGS. **16-17**, Embodiment 4 of the application provides an electronic equipment accessory, which is different from the electronic equipment accessory in the above embodiments mainly in the structure of the rotating device **1**. In this embodiment, the electronic equipment accessory is described with a mobile phone protective case as an example. The electronic equipment accessory comprises an accessory body **2**, a rotating device **1** and a support member **15**. The rotating device **1** and the support member **15** cooperatively form a stand. The accessory body **2** comprises a back panel **21** and a frame **22** connected to the back panel **21**. In this embodiment, the frame **22** is connected to the periphery of the back panel **21** to cooperatively form a receiving cavity **212** for receiving electronic equipment. The electronic equipment is, for example, a mobile phone, a tablet, or other portable electronic terminals. For the sake of a brief description, the electronic equipment is described in the following embodiments with a mobile phone as an example, and correspondingly, the electronic equipment accessory is a mobile phone protective case unit.

(67) Referring to FIGS. **17-19**, the frame **22** comprises a top wall **221** and a bottom wall **222** which are oppositely arranged in a length direction of the accessory body **2**, and generally, the frame **22** further comprises a left sidewall and a right sidewall which are oppositely arranged in a width direction of the accessory body **2**. The back panel **21** comprises a first area **21***a* and a second area **21***b* in the length direction of the accessory body **2**, the first area **21***a* is adjacent to the bottom wall **222**, and the second area **21***b* is located between the first area **21***a* and the top wall **221**. A receiving groove **211** is formed in the first area **21***a* and is located in a side of the back panel **21** opposite to the receiving cavity **212**. A length direction of the receiving groove **211** is parallel to the width direction of the accessory body **2**, in other words, the length direction of the receiving groove **211** is consistent with the width direction of the accessory body **2**. The rotating device **1**, which is mounted to the back panel **211**, comprises a first connecting member **11**, a second connecting

member **12** and a shaft **13**. The support member **15** is connected to the second connecting member **12**. The shaft **13** is inclined with respect to the bottom wall **222**. For example, as shown in FIGS. **16** and **18**, an angle α, which is formed between an axial direction of the shaft **13** and the bottom wall **222**, is an acute angle (for example, the preset acute angle A in other embodiments), and specifically, the angle α is an angle formed between the axial direction of the shaft **13** and the length direction of the bottom wall **222**, i.e., the width direction of the accessory body **2**. In this embodiment, by mounting the rotating device **1** in the first area **21***a* close to the bottom wall **222** of the accessory body **2**, the second area **21***b* of the back panel **21** may have a larger function setting region, in which functional components such as a magsafe magnetic attraction component, a fill-in light, a card case or a ring holder may be installed. In addition, to fulfil a suitable support angle of a stand close to the bottom wall **222** of the accessory body **2**, the shaft **13** of the rotating device **1** is inclined with respect to the bottom wall **222**, for example, when the angle α is 60°, an inclination angle β of the back panel **21** of the accessory body **2** with respect to a support surface is 68° (as shown in FIG. **22**). Preferably, the angle α is greater than or equal to 45° and less than or equal to 75°, such that the inclination angle β of the back panel **21** of the accessory body **2** with respect to the support surface may be greater than or equal to 60° and less than or equal to 75° to adapt to the using habits of most users. It should be noted that the support surface typically refers to a plane cooperatively defined by a support point at an end, away from the back panel **21**, of the support member **15** and another support point on the bottom wall **222** when the support member **15** (the second connecting member **12** which acts as a support member in a case where the support member **15** is omitted) is in a support state with respect to the accessory body **2**; and the support points refer to contact points between the support member **15** and a support surface (such as a surface of a desk) as well as between the bottom wall **222** and the support surface. The support member **15** shown in FIG. **22** can vertically support the accessory body **2** on the support surface. It should be noted that the support member **15** in this embodiment not only can vertically support the accessory body **2**, but also can horizontally support the accessory body **2**, that is, the length direction of the accessory body **2** parallel to the support surface. It should also be noted that, in some embodiments, the receiving groove **211** defined in the first area **21***a* may be omitted, and correspondingly, the support member **15** is protrusively arranged at the first area **21***a*.

(68) In some embodiments, referring to FIGS. **19-22**, the first connecting member **11** is fixedly arranged in the receiving groove **211**, the second connecting member **12** is rotatably connected to the first connecting member **11** by means of the shaft **13**, and the second connecting member **12** is switchable with respect to the first connecting member **11** between a folded state (shown in FIG. **20**) and a support state (shown in FIG. **21**). With reference to FIGS. **19** and **20**, in the folded state, the second connecting member **12** and the support member **15** are at least partly received in the receiving groove **211**. With reference to FIGS. **19**, **21** and **22**, in the support state, the second connecting member **12** and the support member **15** are rotated away from the receiving groove **211**, such that the accessory body **2** can be supported by the support member **15** in an oblique state. In other embodiments, the first connecting member **11** of the rotating device **1** may be omitted, and the second connecting member **12** is fixed to the back panel **21** of the accessory body **2** by means of the shaft **13**. The support member **15** may be omitted, and the accessory body **2** can be supported by the second connecting member **12** which acts as a support member.

(69) In some embodiments, referring to FIGS. **19** and **23-25**, two second rotary connection portions **122** and a limiting portion **124** located between the two second rotary connection portions **122** are arranged at one end of the second connecting member **12**, and the two second rotary connection portions **122** each defines a coupling hole **1220** matching with the shaft **13**. For instance, the shaft **13**, for example, has two opposite flat surfaces and two opposite arc surfaces connected between the two flat surfaces; correspondingly, the coupling hole **1220** has two opposite flat side walls and two opposite arc side walls connected between the two flat side walls, such that the shaft **13** can be mounted and fitted in the coupling holes **1220** to realize a connection between the shaft **13** and the

coupling portions **122** and to allow the second connecting member **12** to be rotatable with the shaft **13**. In addition, a first rotary connection portion **112** is arranged at one end of the first connecting member **11**. The first rotary connection portion **112** comprises a first stop surface **1121** and a sleeving hole **1120** in which the shaft **13** is fitted. For instance, an opening is formed in the circumferential wall of the sleeving hole **1120**. That is, the sleeving hole **1120** is not completely closed in the circumferential direction, which allows the sleeving hole **1120** to expand. When the second connecting member **12** of the rotating device **1** is rotated outwards to be unfolded, the shaft **13** is driven to rotate to thereby drive the first rotary connection portion **112** to expand to some extent and deform elastically. When the second connecting member **12** rotates inwards (in a direction close to the first connecting member **11**) by a certain distance, the elastic restoring force of the first rotary connection portion **112** will drive the shaft **13** to quickly rotate to its final position to thereby make the second connecting member **12** automatically cover the first connecting member **11**. That is, the shaft **13** is rotatably fitted in the first rotary connection portion **112**. In addition, it should be noted that after the rotating device **1** is mounted in the receiving groove **211**, the first connecting member **11** remains stationary with respect to the receiving groove **211** when the second connecting member **12** rotates into or away from the receiving groove **211**. The first connecting member **11** may be secured in the receiving groove **211** with an adhesive or in other suitable ways as long as the first connecting member **11** can be fixed in the receiving groove **211**.
(70) Based on the above description, the shaft **13** is mounted in the coupling holes **1220** and the sleeving hole **1120** and the first rotary connection portion **112** is located between the two second rotary connection portions **122**, such that the second connecting member **12** and the first connecting member **11** are rotatably connected to each other. The limiting portion **124** cooperates with the first stop surface **1121** to define the maximum rotation angle of the second connecting member **12** with respect to the first connecting member **11**, that is, the second connecting member **12** comes in contact with the first stop surface **1121** when rotating to be unfolded to the maximum angle, such that the second connecting member **12** cannot be further unfolded anymore.
(71) In some embodiments, as shown in FIG. **23**, the first connecting member **11** comprises a first connection portion **111** and an extension portion **113** connected between the first rotary connection portion **112** and the first connection portion **111**. Reinforcing ribs **1131** are arranged on the extension portion **113**. The reinforcing ribs **1131** can effectively prevent deformation of the extension portion **113**.
(72) In some embodiments, referring to FIGS. **18** and **19**, a through-opening **215** penetrating through the back panel **21** is formed in an end, close to the shaft **13**, of the receiving groove **211** to function as an avoidance space. The through-opening **215** is arranged to better prevent interference and friction between the second connecting member **12** and the back panel **21** when the second connecting member **12** rotates.
(73) In some embodiments, as shown in FIGS. **19** and **20**, the support member **15** is fixedly arranged on a side, away from the first connecting member **11**, of the second connecting member **12**. The support member **15** can improve the overall aesthetics of the electronic equipment accessory. In addition, to ensure that the electronic equipment accessory can be comfortably held, a side, away from the second connecting member **12**, of the support member **15** will not protrude out of the receiving groove **211** and is preferably flush with the opening end of the receiving groove **211** when the rotating device **1** is in the folded state.
(74) In some embodiments, referring to FIGS. **21** and **23**, the second connecting member **12** comprises a first edge **12*a*** away from the shaft **13** and a second edge **12*b*** adjacent to the first edge **12*a***, wherein the second edge **12*b*** is adjacent to the bottom wall **222** of the frame **22**, and an arc edge **12*c*** is arranged at a joint between the first edge **12*a*** and the second edge **12*b***. By means of the arc edge **12*c***, the electronic equipment accessory can be supported on an uneven surface, that is, any one of points on the arc edge **12*c*** may function as a support point in contacting with a support surface. In other embodiments, the first edge **12*a*** may be configured as an arc edge to replace the

arc edge **12***c* arranged between the first edge **12***a* and the second edge **12***b*.
(75) In some embodiments, as shown in FIG. **17**, a functional component such as a magnetic attraction component **23** is arranged in the second area **21***b* of the back panel **21**, and other functional components such as a fill-in light, a card case and a finger ring grip may also be arranged in the second area **21***b* of the back panel **21**. In addition, it can be understood that, a camera hole is generally also formed in the second area **21***b* to expose a mobile phone camera when the mobile phone is received in the receiving cavity **212** of the accessory body **2**. Further, as shown in FIGS. **17** and **26**, a mounting groove **216** is formed in the second area **21***b* of the back panel **21**. The magnetic component **23** comprises a magnet **231** and a magnet cover **232**, the magnet **231** is received in the mounting groove **216**, and the magnet cover **232** covers the mounting groove **216** to fix the magnet **231** between the mounting groove **216** and the magnet cover **232**. Specifically, the mounting groove **21** comprises a ring-shaped mounting groove **216***a* and a strip-shaped mounting groove **216***b* spaced apart from the ring-shaped mounting groove **216***a*, the magnet **231** comprises a ring-shaped magnet **231***a* and an alignment magnet **231***b*, the magnet cover **232** comprises a ring-shaped cover **232***a* and a strip-shaped cover **232***b*, the ring-shaped magnet **231***a* is received in the ring-shaped mounting groove **216***a*, the ring-shaped cover **232***a* covers the ring-shaped mounting groove **216***a* to fix the ring-shaped magnet **231***a* between the ring-shaped mounting groove **216***a* and the ring-shaped cover **232***a*, the alignment magnet **231***b* is received in the strip-shaped mounting groove **216***b*, and the strip-shaped cover **232***a* covers the strip-shaped mounting groove **216***b* to fix the alignment magnet **231***b* between the strip-shaped mounting groove **216***b* and the strip-shaped cover **232***b*. Typically, a groove for receiving the magnet **231** is formed in a back side of the magnet cover **232**, the ring-shaped magnet **231***a* is configured for attracting an external accessory and generally formed by a plurality of segmented magnets which are arranged at interval in a ring shape, and the alignment magnet **231***b* is a magnet used for alignment. Furthermore, the second area **21***b* of the back panel **21** may be transparent, such that the aesthetics can be improved, and the position of the magnet **231** can be displayed to allow users to quickly align the magnet **231***b* and an accessory to be attracted.
(76) In addition, it should be noted that, in other embodiments, the rotating device **1** provided by this embodiment of the application, as an independent component, may be directly fixed to the back side of electronic equipment such as the back side of a mobile phone to function as a stand, such that the mobile phone can be supported on a support surface obliquely, vertically or horizontally. In this application scenario, referring to FIGS. **23-25**, the rotating device **1** comprises the first connecting member **11**, the second connecting member **12** and the shaft **13**, and further comprises the support member **15**. The first connecting member **11** is fixedly connected to the back side of the electronic equipment, for example, with an adhesive or in other ways to realize a fixed connection between the rotating device **1** and the electronic equipment. The second connecting member **12** is rotatably connected to the first connecting member **11** by means of the shaft **13** such that the second connecting member **12** is rotatable relative to the first connecting member **11** about a rotation axis defined by the axis of the shaft **13**. The support member **15** is fixedly arranged on a side, away from the first connecting member **11**, of the second connecting member **12**, an angle α formed between an axial direction of the shaft **13** and a length direction of the rotating device **1** is an acute angle and is preferably an acute angle which is greater than or equal to 45° and less than or equal to 75°. Further, an end, close to the shaft **13**, of the first connecting member **11** is provided with, for example, a first rotary connection portion **112**, and the first rotary connection portion **112** has a first stop surface **1121** and a sleeving hole **1120** in running fit with the shaft **13**. An end, close to the shaft **13**, of the second connecting member **12** is provided with, for example, two second rotary connection portions **122** and a limiting portion **124** located between the two second rotary connection portions **122**. The two second rotary connection portions **122** each defines a coupling hole **1220** matching with the shaft **13**. The shaft **13** is mounted in the coupling holes **1220** and the sleeving hole **1120** and the first rotary connection portion **112** is located between the two second

rotary connection portions **122** to realize a rotatable connection between the second connecting member **12** and the first connecting member **11**. The limiting portion **124** cooperates with the first stop surface **1121** to define the maximum rotation angle of the second connecting member **12** with respect to the first connecting member **11**. In some embodiments, the second connecting member **12** comprises a first edge **12***a* away from the shaft **13** and a second edge **12***b* adjacent to the first edge **12***a*. An arc edge **12***c* is arranged at a joint between the first edge **12***a* and the second edge **12***b*. The electronic equipment accessory can be supported on an uneven support surface by means of the arc edge **12***c*, that is, any one of points in the arc edge **12***c* may function as a support point. Alternatively, the first arc edge **12***a* is an arc edge which can function as a support edge to replace the arc edge **12***c* arranged between the first edge **12***a* and the second edge **12***b*.

Embodiment 5

(77) Referring to FIGS. **27-29**, Embodiment 5 of the application provides a rotating device **1** and a mobile phone protective case with the rotating device **1**. This embodiment is different from the above embodiments mainly in the structure of the rotating device **1**. In this embodiment, the rotating device **1** is formed as a stand. The rotating device **1** of the mobile phone protective case comprises a first connecting member **11** and a second connecting member **12**. The first connecting member **11** is fixedly secured to the mobile phone protective case. One end of the second connecting member **12** is rotatably connected to the first connecting member **11**. The second connecting member **12** is rotatable between a folded state and an extended state. In the folded state, the second connecting member **12** is stacked on the first connecting member **11**. In the extended state, the second connecting member **12** is rotated away from the first connecting member **11** and an included angle is formed between the second connecting member **12** and the mobile phone protective case such that the second connecting member **12** can act as a support member to support the mobile phone protective case together with a mobile phone mounted in the mobile phone protective case on a support surface. The first connecting member **11** comprises the first connection surface **1111** which is connected to the mobile phone protective case. The first connection surface **1111** is a surface of the first connecting member **11** which is orientated toward the second connecting member **12** when the rotating device **1** is in the folded state.

(78) Compared with the prior art, one end of the second connecting member **12** is rotatably connected to the first connecting member **11**, such that the second connecting member **12** may support the mobile phone protective case at different angles by means of the rotatable connection. When the rotating device **1** is in the folded state, the second connecting member **12** is stacked on the first connecting member **11**, the first connection surface **1111** is the surface, facing the second connecting member **12**, of the first connecting member **11** (when the rotating device **1** is in the folded state), the mobile phone protective case is connected to the first connecting member **11** by means of the first connection surface **1111**, and the first connecting member **11** can be inserted into the interior of the mobile phone protective case to realize connection, which is different from an existing configuration that a stand is connected to an outer surface of the mobile phone protective case. Thus, the second connecting member **12** of the rotating device **1** can be tightly and firmly connected to the mobile phone protective case and is unlikely to fall from the mobile phone protective case. The reliability of the second connecting member **12** acting as a support member and the mobile phone protective case is improved. In addition, the rotating device **1** may be used for supporting a portable power supply, a charge, a mobile phone, a tablet, a reader, a display, an earphone case, and other electronic equipment or electronic equipment accessories.

(79) Referring to FIGS. **28-30**, according to the rotating device **1** in this embodiment of the application, the first connecting member **11** comprises the first connection portion **111**, and the first connection surface **1111** is a surface, orientated toward the second connecting member **12**, of the first connection portion **111** when the rotating device is in the folded state. Specifically, the first connection surface **1111** is arranged on the first connection portion **111**, and the first connection portion **111** is connected to the electronic equipment accessory for example a mobile phone

protective case by means of the first connection surface **111**, and the first connection portion **111** of the first connecting member **11** can be inserted into the interior of the electronic equipment accessory to realize connection such that the second connecting member **12** can be tightly and firmly connected to the electronic equipment accessory and is unlikely to fall from the external electronic equipment, thus improving the reliability of the second connecting member **12** acting as a support member and the electronic equipment accessory.

(80) The first connection surface **1111** may be connected to the electronic equipment accessory by at least one of adhesive connection, magnetic attraction connection and clamping. Specifically, in a case where the first connection surface **1111** of the first connecting member **11** is adhesively connected to the electronic equipment accessory, the adhesive force applied to the first connection surface **1111** of the first connecting member **11** is uniform, and the first connection surface **1111** can be firmly adhered to the electronic equipment accessory. The adhesive connection can be implemented easily and quickly and has a wide application range. In a case where the first connection surface **1111** of the first connecting member **11** is magnetically connected to the electronic equipment accessory, the first connection surface **1111** of the first connecting member **11** is tightly connected to the electronic equipment accessory by magnetic attraction. If the first connection surface **1111** of the first connecting member **11** needs to be disconnected from the electronic equipment accessory, the first connection surface **1111** of the first connecting member **11** can be removed away from the electronic equipment accessory by overcoming the magnetic attraction. By means of a magnetic attraction connection way, connection and disconnection are simple, and the connection efficiency and reliability are improved. In a case where the first connection surface **1111** of the first connecting member **11** is clamped in the electronic equipment accessory, the first connection surface **1111** of the first connecting member **11** is clamped/locked in the electronic equipment accessory, and in this case, quick assembly and disassembly can be realized, and the installation is convenient.

(81) Referring to FIGS. **28-30**, in the case where the first connection surface **1111** is adhesively connected to the electronic equipment accessory, the adhesive connection area between the first connection surface **1111** and the electronic equipment accessory is greater than half of the area of the first connecting member **11**. Generally, the adhesive connection area can bear a load. A larger adhesive connection area can bear a larger load. The adhesive connection area between the first connection surface **1111** and the electronic equipment accessory is set to be greater than half of the area of the first connecting member **11**, such that a larger load can be borne by the adhesive connection area between the first connection surface **1111** and the electronic equipment accessory, and the first connection surface **1111** and the electronic equipment accessory can be adhesively connected more firmly.

(82) Further, the first connecting member **11** further comprises an extension portion **113** and a bent portion **114** located between the extension portion **113** and the first connection portion **111**, and one end of the second connecting member **12** is rotatably connected to the extension portion **113**. The first connecting member **11** comprises a connection section contact surface **1132** which is a surface, opposite to the second connecting member **12**, of the extension portion **113** when the rotating device **1** is in the folded state. The connection section contact surface **1132** is configured to be connected to an electronic equipment accessory. Specifically, when the second connecting member **12** is in the extended state to allow the electronic equipment accessory together with the second connecting member **12** to support an electronic equipment on a support surface, the second connecting member **12** provides a support force for the extension portion **113** to realize tight connection between the connection section contact surface **1132** of the extension portion **113** and the electronic equipment accessory; and the bent portion **114** provides an opposite force for the first connection portion **111** due to the principle of leverage to realize a tight connection between the first connection surface **1111** of the first connection portion **111** and the electronic equipment accessory. That is, the first connection surface **1111** and the connection section contact surface **1132**

respectively abut against the electronic equipment accessory from two opposing directions when the second connecting member **12** is in the extended state to support the electronic equipment accessory on the support surface. By means of the tight connection between the first connection surface **1111** and the electronic equipment accessory and the tight connection between the connection section contact surface **1132** and the electronic equipment accessory in two opposing directions, the first connecting member **11** can be connected to the electronic equipment accessory more firmly, and the force acted on the electronic equipment accessory is distributed more uniformly when the second connecting member **12** supports the first connecting member **11** and the electronic equipment accessory, such that the rotating device **1** is unlikely to fall from the electronic equipment accessory.

(83) Further, reinforcing ribs **1131** are arranged on a surface, opposite to the connection section contact surface **1132**, of the extension portion **113** to improve the deformation resistance of the extension portion **113**. Referring to FIGS. **28-30**, the area of the first connection surface **1111** is greater than or equal to the area of the connection section contact surface **1132**. Specifically, the first connection portion **111** of the first connecting member **11** is inserted into the interior of the electronic equipment accessory and the first connecting member **11** is connected to the electronic equipment accessory by means of the first connection surface **1111** of the first connection portion **111** which is located inside the electronic equipment accessory. To ensure that the first connecting member **11** can be connected to the electronic equipment accessory firmly and is unlikely to fall away from the electronic equipment accessory, the area of the first connection surface **1111** is set to be greater than the area of the connection section contact surface **1132** to realize an enlarged connection area between the first connecting member **11** and the electronic equipment accessory inside the electronic equipment accessory, such that the first connecting member **11** and the electronic equipment accessory are firmly connected.

(84) An electronic equipment accessory such as a mobile phone protective case in this embodiment of the application comprises an accessory body **2** and the rotating device **1** described above. The accessory body **2** comprises an insertion and connection portion. The first connecting member **11** is inserted into the insertion and connection portion and the first connection surface **1111** is connected to the accessory body **2**. Specifically, one end of the second connecting member **12** is rotatably connected to the first connecting member **11**, the second connecting member **12** can support the electronic equipment accessory at different angles by means of the rotatable connection with the first connecting member **11**. The first connecting member **11** of the rotating device **1** is inserted into the insertion and connection portion of the accessory body **2**, and the first connection surface **1111** is connected to the accessory body **2**. The first connecting member **11** is inserted into the accessory body **2** to realize connection, which is different from existing solutions where a stand is connected to an outer surface of the accessory body **2**. In the present application, the rotating device **1** can be tightly connected to the accessory body **2** and is unlikely to fall from the accessory body **2**, thus improving the reliability of the rotating device **1** and the electronic equipment accessory.

(85) Referring to FIGS. **28-29**, according to the electronic equipment accessory in this embodiment of the application, the accessory body **2** comprises a back panel **21**, the insertion and connection portion (not labelled) is arranged on the back panel **21** and comprises a through-hole **217** which passes through an inner surface **218***a* and an outer surface **218***b* of the back panel **21**. Specifically, one end of the second connecting member **12** is rotatably connected to the first connecting member **11**. Referring to FIGS. **32** and **33**, the second connecting member **12** can support the electronic equipment accessory at different angles by means of the rotatable connection with the first connecting member **11**. The first connecting member **11** of the rotating device **1** is inserted into the interior side of the back panel **21** via the through-hole **217** of the back panel **21**, such that the rotating device **1** is tightly connected to the back panel **21** and is unlikely to fall from the back panel **21**, thus improving the reliability of the rotating device **1** and the back panel **21**. The electronic equipment accessory is detachably mounted on a back side of an electronic equipment

such as a mobile phone. When the electronic equipment accessory is mounted on the electronic equipment, the first connecting member **11** is sandwiched between the back panel **21** and the electronic equipment, such that the rotating device **1** is connected to the back panel **21** more tightly and is unlikely to fall from the back panel **21**.

(86) Referring to FIGS. **28-30**, the first connecting member **11** comprises the first connection portion **111**, the bent portion **114** and the extension portion **113** which are arranged in sequence. The first connecting member **11** is inserted into the through-hole **217**, the extension portion **113** is located on one side of the outer surface **218***b* of the back panel **21**, the first connection portion **111** is located on one side of the inner surface **218***b* of the back panel **21**, and the bent portion **114** is located at the through-hole **217**. Specifically, during the process of the rotating device **1** being inserted into and connected to the back panel **21**, the first connection portion **111** of the first connecting member **11** is inserted into the through-hole **217**, and when the bent portion **114** reaches the through-hole **217**, the first connecting member **11** is rotated with the through-hole **217** as a rotation center until the first connection portion **111** abuts against the inner surface **218***a* of the back panel **21**, and at this moment, the extension portion **113** abuts against the outer surface **218***b* of the back panel **21**. When the second connecting member **12** is in the extended state to support an electronic equipment accessory on a support surface, the second connecting member **12** provides a support force to urge the extension portion **113** to abut against the back panel **21**; and the bent portion **114** provides an opposite force for the first connection portion **111** due to the principle of leverage to urge the first connection portion **111** to abut against the back panel **21**. Thus, the first connecting member **11** and the back panel **21** are connected more firmly, and the force acted on the back panel **21** is distributed more uniformly when the second connecting member **12** is in the extended state to support the electronic equipment accessory on the support surface. In addition, the bent portion **114** is used to mount the rotating device **1** to the back panel **21**, such that when the rotating device **1** needs to be moved away from the back panel **21**, the first connecting member **11** needs to be rotated with the through-hole **217** as the rotation center to make the bent portion **114** away from the through-hole **217**, then the first connection portion **111** can be pulled out of the through-hole **217**. If the above-described operation is not performed, the rotating device **1** cannot be directly pulled out of the back panel **21**, such that the rotating device **1** is less likely to fall from the back panel **21**. Since the rotating device **1** is mounted on the back panel **21**, users just need to install the back panel **21** on electronic equipment to thereby mount the rotating device **1** to the electronic equipment and can use the rotating device **1** anytime and anywhere according to their requirements, thus improving user experience.

(87) Referring to FIGS. **28-30**, the first connection surface **1111** is connected to the inner surface **218***a* of the back panel **21**. Specifically, when the first connecting member **11** of the rotating device **1** is inserted into the through-hole **217** of the back panel **21**, that is, when the first connecting member **11** is inserted into the interior of the back panel **21**, the first connection surface **1111** of the first connecting member **11** is connected to the inner surface **218***a* of the back panel **21**, such that the rotating device **1** is tightly connected to the back panel **21** and is unlikely to fall from the back panel **21**, thus improving the reliability of the rotating device **1** and the back panel **21**.

(88) Referring to FIGS. **28** and **29**, the electronic equipment accessory further comprises a first adhesive part **115** arranged between the first connection surface **1111** and the inner surface **218***a* of the back panel **21**, and the first connection surface **1111** is adhesively connected to the inner surface **218***a* of the back panel **21** by means of the first adhesive part **115**. By means of the adhesive part **115** arranged between the first connection surface **1111** and the inner surface **218***a* of the back panel **21**, the first connection surface **1111** can be adhesively connected to the inner surface **218***a* of the back panel **21**. Specifically, the first adhesive part **115** is an adhesive part with adhesive properties on both sides, such as a glue, a double-sided adhesive tape or a nano-glue. By use of the first adhesive part **115**, the first connection surface **1111** can be more tightly connected to the inner surface **218***a* of the back panel **21**, such that the rotating device **11** can be connected to the back

panel **21** more firmly and the rotating device **1** is less likely to fall from the back panel **21**.
(89) Further, the electronic equipment accessory further comprises a magnetic attraction component such as the magnetic component **23** described in other embodiments. The first connecting member **11** is made from a magnet conductive metal or a permanent magnet. The magnetic attraction component is arranged at the inner surface **218***a* of the back panel **21**, and the first connection surface **1111** is magnetically connected to the inner surface **218***a* of the back panel **21** by means of the magnetic attraction component. Specifically, the magnetic attraction component is arranged on the inner surface **218***a* of the back panel **21**, the first connecting member **11** is made from a magnet conductive metal, and the magnetic component magnetically attracts the first connection surface **1111** to cause the first connection surface **1111** to be tightly connected to the inner surface **218***a* of the back panel **21**, such that the first connecting member **11** is more firmly connected to the back panel **21**, and the rotating device **1** is less likely to fall from the back panel **21**.
(90) Referring to FIGS. **28** and **29**, the back panel **21** comprises a through-opening **215**, and a rotary joint between the first connecting member **11** and the second connecting member **12** is located in the through-opening **215**. Specifically, the second connecting member **12** needs to be frequently adjusted/rotated around the rotary joint between the first connecting member **11** and the second connecting member **123** in order to support the electronic equipment accessory at different angles, and the first connection surface **1111** of the first connecting member **11** is connected to the inner surface **218***a* of the back panel **21**, that is, one portion of the first connecting member **11** is inserted into the interior of the back panel **21**, and the other portion of the first connecting member **11** is located on the outer surface **218***b* of the back panel **21**, in other words, the rotary joint between the first connecting member **11** and the second connecting member **12** is located on the outer surface **218***b* of the back panel **21**, so when the second connecting member **12** is rotated with respect to the first connecting member **11**, friction will be generated between the rotary joint between the first connecting member **11** and the second connecting member **12** and the outer surface **218***b* of the back panel **21**. In order to protect the rotary joint between the first connecting member **11** and the second connecting member **12** from being damaged by the friction, the through-opening **215** is formed to receive the rotary joint between the first connecting member **11** and the second connecting member **12**, so as to reduce the wear of the rotary joint between the first connecting member **11** and the second connecting member **12**.
(91) The electronic equipment accessory further comprises a second adhesive part **116**, the first connecting member **11** further comprises a connection section contact surface **1132** which is a surface, opposite to the second connecting member **12**, of the extension portion **113** when the rotating device **1** is in the folded state. The second adhesive part **116** is arranged between the connection section contact surface **1132** and the outer surface **218***b* of the back panel **21**. Specifically, the first connection surface **1111** of the first connecting member **11** is connected to the inner surface **218***a* of the back panel **21**, that is, one portion of the first connecting member **11** is inserted into the interior of the back panel **21** and the other portion of the first connecting member **11** is located on the outer surface **218***b* of the back panel **21** and is adhesively connected to the outer surface **218***b* of the back panel **21** by means of the second adhesive part **116**, in other words, the second adhesive part **116** is arranged between the connection section contact surface **1132** and the outer surface **218***b* of the back panel **21**, and the extension portion **113** is adhesively connected to the outer surface **218***b* of the back panel **21** by means of the second adhesive part **116**. Specifically, the second adhesive part **16** is an adhesive part with adhesive properties on both sides, such as a glue, a double-sided adhesive tape or a nano-glue. By use of the second adhesive part **116**, the first connecting member **11** can be more tightly connected to the outer surface **218***b* of the back panel **21**, such that the first connecting member **11** is connected to the back panel **21** more firmly, and the rotating device **1** is less likely to fall from the back panel **21**.
(92) Referring to FIGS. **28** and **29**, the electronic equipment accessory comprises a first attachment

part **24** which covers the first connecting member **11** and the insertion and connection portion of the back panel **21** for preventing the first connecting member **11** from being pulled out of the back panel **21**. Specifically, the first attachment part **24** covers the first connecting member **11** and the insertion and connection portion of the back panel **21** and blocks the first connecting member **11** from being pulled out of the back panel **21**, such that when users hold the second connecting member **12** by hands, the first attachment part **24** provides a resistance to prevent the first connecting member **11** from being pulled out of the back panel **21**, thus further improving the connection firmness between the rotating device **1** and the back panel **21**.

(93) Referring to FIGS. **27** and **28**, a receiving groove **211** is formed in the outer surface **218***b* of the back panel **21**, and the through-hole **217** is located in the receiving groove **211**. Specifically, the receiving groove **211** is concavely formed in the outer surface **218***b* of the back panel **21** and used for receiving the rotating device **1**. Referring to FIG. **27**, the first connecting member **11** of the rotating device **1** is inserted into the insertion and connection portion of the back panel **21**, the second connecting member **12** of the rotating device **1** is stacked on the first connecting member **11***b* when not used, and at this moment, the second connecting member **12** is received in the receiving groove **211**, that is, the rotating device **1** is received in the receiving groove **211** and is exactly flush with the back panel **21**, such that the back panel **21** can be flatly placed on a support surface such as a top surface of a desk.

(94) Referring to FIG. **27**, the support member of the rotating device **1** comprises an arcuate outer end away from the shaft. The receiving groove **211** comprises an arcuate inner end wall away from the shaft, the receiving groove **211** defines a length direction parallel to the width direction of the back panel **21** and a width direction parallel to the length direction of the back panel **21**, and the arcuate inner end wall of the receiving groove **211** faces the arcuate outer end of the support member with a gap formed therebetween. The gap has a size S in the width direction of the back panel **21**. The size S refers to the distance between the outmost edge of the support member and the innermost edge of the inner end wall of the receiving groove **211**. The arcuate inner end wall of the receiving groove **211** away from the shaft comprises an inclined surface M which is inclined outwardly in a direction from the inner surface toward the outer surface of the back panel **21**. The projection of the inclined surface M on the inner surface of the back panel **21** in the thickness direction of the back panel **21** has a size, in the width direction of the back panel **21**, which gradually decreases from a middle of the inclined surface M to opposite sides of the inclined surface M in the length direction of the back panel **21**. Thus, the size S of the gap in the length direction of the receiving groove **211** gradually decreases from a middle of the gap to opposite sides of the gap in the width direction of the receiving groove **211**, which facilitates users' fingers to enter into the middle of the gap to open the support member. Preferably, the size S of the gap has a maximum value at the middle of the gap and the maximum value is in a range of 1-2.5 mm. The projection of the inclined surface M on the inner surface of the back panel **21** in the thickness direction of the back panel **21** has a maximum value at the middle of projection of the inclined surface M and the maximum value of the projection of the inclined surface M is in a range of 1.5-3.5 mm. The support member has a width in a range of 18-25 mm and a length in a range of 65-75 mm. The thickness, in the thickness direction of the back panel **21**, of the end of the support member away from the shaft is less than that of the middle portion of the support member. The length of the support member refers to the distance from the leftmost end of the support member to the rightmost end of the support member as shown in FIG. **27** where the second connecting member **12** acts as the support member.

(95) The electronic equipment accessory further comprises a second attachment part **25** which covers the bottom of the receiving groove **211** from the interior side of the back panel **21**, and/or the attachment part **25** covers the inner surface **218***a* of the back panel **21** and the first connecting member **11**. Specifically, the back panel **21** defines the through-hole **217** and one portion of the first connecting member **11** is inserted through the through-hole **217** to allow the first connection

surface **1111** to be connected to the inner surface **218***a* of the back panel **21**. In order to prevent the portion, inserted into the back panel **21**, of the first connecting member **11** from contacting with electronic equipment such as a mobile phone or a tablet mounted inside of the back panel **21**, the second attachment part **25** is arranged to cover the portion, inserted into the back panel **21**, of the first connecting member **11**. The second attachment part **25** can withstand the support force of the second connecting member **12** when the second connecting member **12** supports the back panel **21** and the electronic equipment such as the mobile phone and the tablet in the back panel **21**, such that imprints will not be left on the electronic equipment when the electronic equipment is supported for a long time, thus guaranteeing the aesthetics of the accessory body.

(96) It can be understood that the above embodiments are merely used for explaining the application by way of examples, and the technical solutions of the embodiments can be combined freely and used together without causing conflicts and contradictions of the technical features and deviating from the purpose of the application.

(97) Finally, it should be noted that the above embodiments are merely used for explaining the technical solutions of the application rather than limiting the application. Although the application has been described in detail with reference to the above embodiments, those ordinarily skilled in the art should understand that modifications can be made to the technical solutions recorded in the above embodiments or equivalent substitutions can be made to part of technical features in the above embodiments, and all these modifications or substitutions will not make the essence of the corresponding technical solutions deviate from the spirit and scope of the technical solutions of the embodiments of the application.

## Claims

1. An electronic equipment accessory, comprising: an accessory body for attaching to an electronic device, the accessory body comprising a back panel, the back panel having a top edge and a bottom edge opposite to the top edge, the back panel comprising a first area adjacent to the bottom edge; and a rotating device mounted to the back panel and located at the first area, the rotating device comprising a shaft which inclines with respect to the bottom edge; wherein the rotating device comprises a first connecting member and a second connecting member which are pivotably connected to each other via the shaft; the back panel defines a through opening in which the shaft is located, the through opening passing through the back panel in a thickness direction of the back panel; the back panel comprises an inner surface and an outer surface opposite to the inner surface in the thickness direction of the back panel; a receiving groove is defined in the outer surface of the back panel and the second connecting member is received in the receiving groove; the second connecting member acts as or is connected to a support member; the support member comprises an arcuate outer end opposite to the shaft; the receiving groove comprises an arcuate inner end wall opposite to the shaft; the arcuate inner end wall of the receiving groove comprises an inclined surface, and a projection of the inclined surface on the inner surface of the back panel in the thickness direction of the back panel comprises an arcuate outer edge and an arcuate inner edge; a gap is formed between the arcuate inner edge and a projection of the arcuate outer end of the support member on the inner surface of the back panel in the thickness direction of the back panel; the receiving groove defines a length direction parallel to the bottom edge of the back panel and a width direction; and a size of the gap in the length direction gradually decreases from a middle of the gap to opposite sides of the gap in the width direction.
2. The electronic equipment accessory according to claim 1, wherein an angle formed between an axial direction of the shaft and a length direction of the bottom edge is greater than or equal to 45° and less than or equal to 75°.
3. The electronic equipment accessory according to claim 1, wherein the second connecting member is pivotable relative to the first connection member between an initiate position in which

the first connecting member and the second connecting member are located at the first area and distributed on opposite sides of the through opening in a width direction of the back panel and an inclined position in which the second connecting member is inclined relative to the first connecting member; and the width direction of the back panel is perpendicular to the thickness direction of the back panel.
4. The electronic equipment accessory according to claim 3, wherein a support member is connected to the second connecting member and rotatable together with the second connecting member relative to the first connecting member, and the first connecting member is fixedly connected to the back panel; the support member is connected to a side of the second connecting member opposite to the back panel, the second connecting member is connected to the shaft.
5. The electronic equipment accessory according to claim 1, wherein the back panel comprises a second area which is located between the first area and the top edge, and a magnetic attraction component is disposed at the second area.
6. The electronic equipment accessory according to claim 5, wherein the magnetic attraction component comprises a first magnetic attraction member which comprises a plurality of segmented magnets arranged in a ring shape; and the back panel is provided with a ring-shaped cover stacked on the first magnetic attraction member in a thickness direction of the back panel.
7. The electronic equipment accessory according to claim 6, wherein the magnetic attraction component further comprises a second magnetic attraction member which has a strip shape and is located between the first magnetic attraction member and the rotating device; and the back panel is provided with a strip-shaped cover with two arcuate ends, and the strip-shaped cover is stacked on the two sections of the second magnetic attraction member in the thickness direction of the back panel.
8. An electronic equipment accessory comprising: an accessory body for attaching to an electronic device, the accessory body comprising back panel, the back panel having a top edge and a bottom edge opposite to the top edge, the back panel comprising a first area adjacent to the bottom edge; and a rotating device mounted to the back panel and located at the first area, the rotating device comprising a shaft which inclines with respect to the bottom edge; wherein the rotating device comprises a first connecting member and a second connecting member which are pivotably connected to each other via the shaft; the back panel defines a through opening in which the shaft is located, the through opening passing through the back panel in a thickness direction of the back panel; wherein the back panel comprises an inner surface and an outer surface opposite to the inner surface in the thickness direction of the back panel; a receiving groove is defined in the outer surface for receiving the second connecting member; at least one part of the first connecting member is fixed to the inner surface of the back panel opposite to the receiving groove such that a part of the back panel is located between the at least one part of the first connecting member and the second connecting member in the thickness direction of the back panel; and at least another part of the first connecting member extends outwardly to pass through the through opening to connect the second connecting member via the shaft, or at least another part of the first connecting member is received in the through opening to connect the second connecting member via the shaft.
9. The electronic equipment accessory according to claim 8, wherein the receiving groove comprises a first bottom wall and a second bottom wall offset from the first bottom wall in the thickness direction of the back panel, a through hole is defined between the first bottom wall and the second bottom wall, and the first connecting member passes through the through hole.
10. The electronic equipment accessory according to claim 9, wherein the first connecting member comprises a first connection portion, a bent portion and an extension portion arranged in sequence; the extension portion is located at the outer surface of the back panel; the first connection portion is located at the inner surface of the back panel; and the bent portion is located at the through hole.
11. The electronic equipment accessory according to claim 8, further comprising a first attachment part which is located at the receiving groove and covers an outside of the first connecting member,

wherein the first attachment part has a periphery conformed to that of the first connecting member.
12. The electronic equipment accessory according to claim 8, further comprising an attachment part which is located at the inner surface of the back panel opposite to the receiving groove; and the attachment part completely covers the first connecting member and the through opening.
13. The electronic equipment accessory according to claim 8, wherein the receiving groove comprises a first bottom wall and a second bottom wall offset from the first bottom wall in the thickness direction of the back panel; a through hole is defined between the first bottom wall and the second bottom wall; and the first bottom wall is located between the through hole and the through opening, one part of the first connecting member is located outside of the first bottom wall while another part of the first connector member is located inside of the second bottom wall in the thickness direction of the back panel.
14. An electronic equipment accessory, comprising: an accessory body comprising a back panel, the back panel having a top edge and a bottom edge opposite to the top edge, the back panel comprising a first area adjacent to the bottom edge; and a rotating device mounted to the back panel and located at the first area, the rotating device comprising a shaft which inclines with respect to the bottom edge; wherein the rotating device comprises a first connecting member and a second connecting member which are pivotably connected to each other via the shaft; and the back panel defines a through opening in which the shaft is located; the back panel comprises an inner surface and an outer surface opposite to the inner surface in a thickness direction of the back panel; a receiving groove is defined in the outer surface for receiving the rotating device; at least one part of the first connecting member is fixed to the inner surface of the back panel; the receiving groove comprises a first bottom wall and a second bottom wall offset from the first bottom wall in the thickness direction of the back panel; the through opening is defined between the first bottom wall and the second bottom wall; and the first bottom wall is sandwiched between the first connecting member and a mounting member in the thickness direction of the back panel; wherein one side of the mounting member facing the second connecting member defines a notch, and the shaft is received in the notch.
15. The electronic equipment accessory according to claim 14, wherein the second connecting member acts as or is connected to a support member; the support member comprises a support point formed at a free end thereof; when the free end of the support member is moved away from the accessory body to be located at a support state, the support point and the bottom edge cooperatively define a support surface, and an included angle formed between the support surface and the back panel is greater than or equal to 60° and less than or equal to 75°.